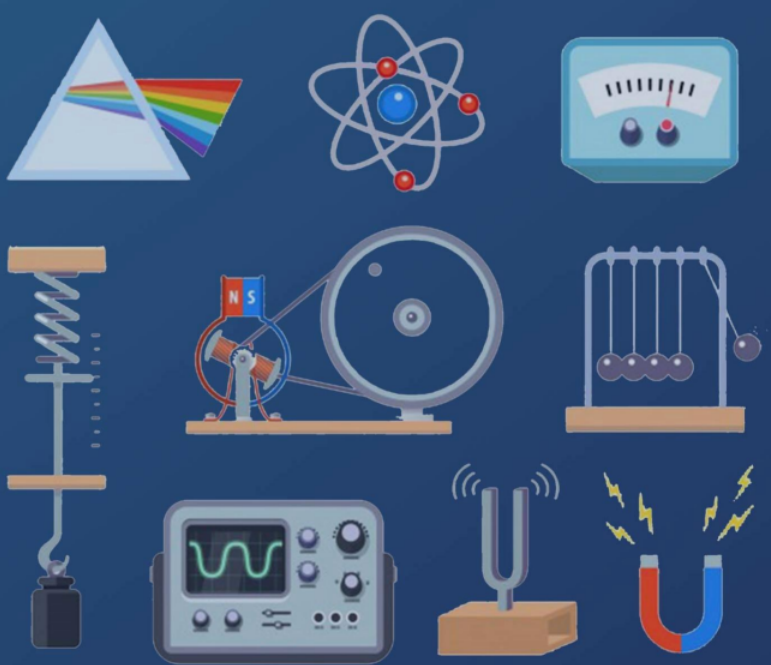
PHYSICS

Heat Transfer



SHEET

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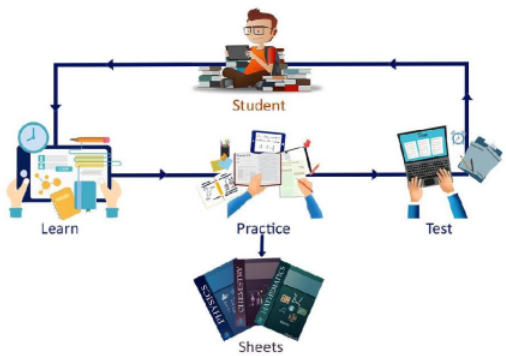
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Modes of Heat Transfer			1	1
2	Conduction-Introduction	1	1	2	
3	Thermal Resistance and Junction Rule	2-4	2,8	3,4	2,18
4	Conduction-Miscelleneous	5,6	3,9	5-7	10-12,19
5	Convection	7	4		3
6	Radiation-Introduction	8-10	5,10	8	4,5
7	Rate of Cooling - Part 1	11	6,11	9,10	
8	Rate of Cooling - Part 2	12	12	11-13	6,7, 13-15,20
9	Wien's displacement law	13,14	7	14-16	8,16,17
10	After complete chapter	15,16			9

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Conduction-Introduction

Q1. Which of the following cylindrical rods will conduct most heat, when their ends are maintained at the same steady temperature:-

(A) length 1 m; radius 1 cm

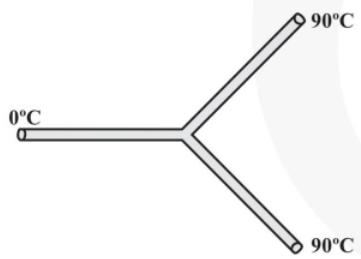
(B) length 2 m; radius 1 cm

(C) length 2 m; radius 2 cm

(D) length 1 m; radius 2 cm

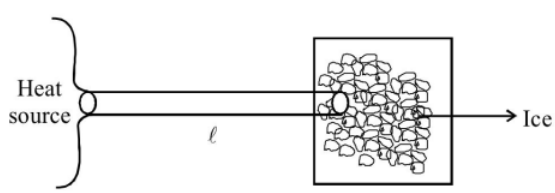
Thermal Resistance and Junction Rule

Q2. Three rods made of the same material and having the same cross–section have been joined as shown in the figure. Each rod is of the same length. The left and right ends are kept at 0°C and 90°C respectively. The temperature of junction of the three rods will be :

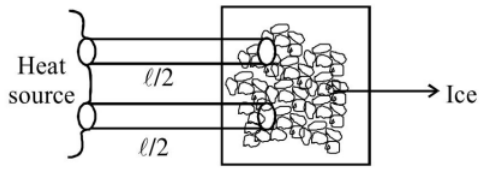


(A) 45°C (B) 60°C (C) 30°C (D) 20°C

Q3. A rod of length 'ℓ' and cross-section 'A' is used to melt a piece of ice as shown.



Now if the rod broken into two equal parts and is arranged as shown.

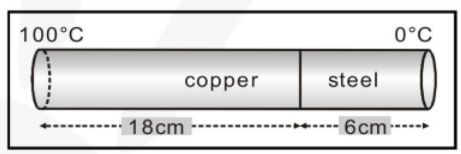


Time taken to melt ice in second use becomes.

(A) Half (B) One-fourth

(C) Twice (D) Four times

Q4. The coefficient of thermal conductivity of copper is nine times that of steel. In the composite cylindrical bar shown in the figure what will be the temperature at the junction of copper and steel ?



(A) 75°C (B) 67°C

(C) 33°C (D) 25°C

Conduction-Miscelleneous

Q5. **Statement–1 :** The bulb of one thermometer is spherical while that of the other is cylindrical. Both have equal amount of mercury. The response of the cylindrical bulb thermometer will be quicker.

Statement–2 : Heat conduction in a body is directly proportional to cross–sectional area.

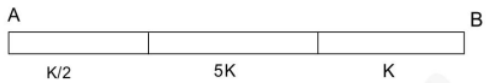
(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

Q6. A composite rod made of three rods of equal length and cross-section as shown in the fig. The thermal conductivities of the materials of the rods are K/2, 5K and K respectively. The end A and end B are at constant temperatures. All heat entering the end A goes out of the end B, there being no loss of heat from the sides of the bar. The effective thermal conductivity of the bar is



(A) 15K/16 (B) 6K/13
(C) 5K/16 (D) 2K/13

Convection

Q7. In which of the following process, convection does not take place primarily :-
(A) Sea and land breeze
(B) Boiling of water
(C) Warming of glass of bulb due to filament
(D) Heating air around a furnace

Radiation-Introduction

Q8. The rectangular surface of area 8cm × 4 cm of a black body at a temperature of 127°C emits energy at the rate of E per second. If the length and breadth of the surface are each reduced to half of the initial value and the temperature is raised to 327°C, the rate of emission of energy will become.

(A) $\frac{3}{8}E$ (B) $\frac{81}{16}E$
(C) $\frac{9}{16}E$ (D) $\frac{81}{64}E$

Q9. The rate of emission of radiation of a black body at 273°C is E, then the rate of emission of radiation of this body at 0°C will be :-

(A) $\frac{E}{16}$ (B) $\frac{E}{4}$ (C) $\frac{E}{8}$ (D) 0

Q10. Two spheres of the same material have radii 1 m and 4 m and temperatures 4000 K and 2000 K respectively. The ratio of the energy radiated per second by the first sphere to that by the second is-
(A) 1 : 1 (B) 16 : 1 (C) 4 : 1 (D) 1 : 9

Rate of Cooling - Part 1

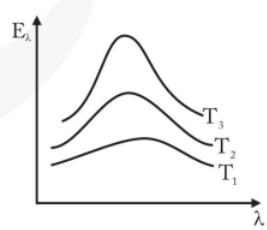
Q11. A black body calorimeter filled with hot water cools from 60°C to 50°C in 4 min and 40°C to 30°C in 8 min. The approximate temperature of surrounding is :
(A) 10°C (B) 15°C (C) 20°C (D) 25°C

Rate of Cooling - Part 2

Q12. A system S receives heat continuously from an electrical heater of power 10W. The temperature of S becomes constant at 50°C when the surrounding temperature is 20°C. After the heater is switched off, S cools from 35.1°C to 34.9°C in 1 minute. The heat capacity of S is
(A) 100J/°C (B) 300J/°C
(C) 750J/°C (D) 1500J/°C

Wien's displacement law

Q13. Variation of radiant energy emitted by sun, filament of tungsten lamp and welding arc as a function of its wavelength is shown in figure. Which of the following option is the correct match :



(A) Sun–T_1, tungsten filament–T_2, welding arc–T_3
(B) Sun–T_2, tungsten filament–T_1, welding arc–T_3
(C) Sun–T_3, tungsten filament–T_2, welding arc–T_1
(D) Sun–T_3, tungsten filament–T_1, welding arc–T_2

Q14. The Wien's displacement law express relation between :-

(A) wavelength corresponding to maximum energy and temperature

(B) radiation energy and wavelength

(C) temperature and wavelength

(D) colour of light and temperature

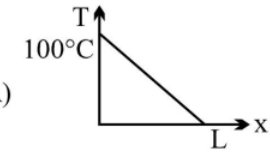
After Full Chapter

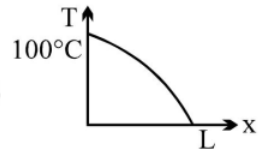
Q15. The dimensional formula for thermal resistance is :-

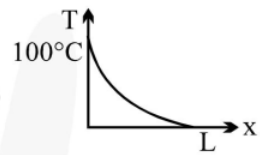
(A) $M^{-1}L^{-2}T^{3}\theta$ (B) $M^{-1}L^{-2}T^{-3}\theta$

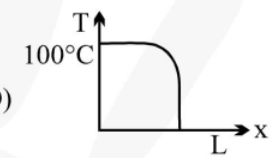
(C) $ML^{2}T^{-2}\theta$ (D) $ML^{2}T^{2}\theta^{-1}$

Q16. A rod of length L and uniform cross-sectional area has varying thermal conductivity which changes linearly from 2K at end A to K at the other end B. The ends A and B of the rod are maintained at constant temperature 100°C and 0°C, respectively. At steady state, the graph of temperature : T = T(x) where x = distance from end A will be

(A)


(B)


(C)


(D)


Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. On a cold morning, a metal surface will feel colder to touch than a wooden surface because :-

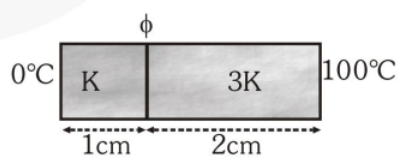
(A) metal has high specific heat

(B) metal has high thermal conductivity

(C) metal has low specific heat

(D) metal has low thermal conductivity

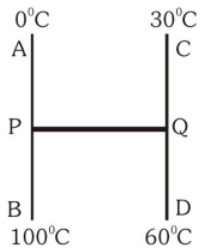
Q2. Two bars of thermal conductivities K and 3K and lengths 1 cm and 2 cm respectively have equal cross-sectional area. They are joined lengths wise as shown in the figure. If the temperature at the ends of this composite bar is 0°C and 100°C respectively (see figure), then the temperature of the interface (ϕ) is :–



(A) 50°C (B) $\frac{100}{3}$°C

(C) 60° C (D) $\frac{200}{3}$ °C

Q3. Three identical rods AB, CD and PQ are joined as shown. P and Q are mid points of AB and CD respectively. Ends A, B, C and D are maintained at 0^0C, 100^0C, 30^0C and 60^0C respectively. The direction of heat flow in PQ is :-



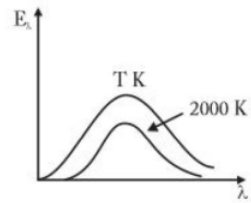
(A) From P to Q
(B) From Q to P
(C) Heat does not flow in PQ
(D) Data not sufficient

Q4. While measuring the thermal conductivity of a liquid, we keep the upper part hot and the lower part cool so that :–
(A) convection may be stopped
(B) radiation may be stopped
(C) heat conduction is easier downwords
(D) it is easier and convenient to do so

Q5. If the absorption coefficient and reflection coefficient of a surface of a body are 0.4 and 0.6 respectively then:–
(A) Emissive power will be 0.2
(B) Transmission power will be 0.2
(C) Body will be totally transparent
(D) Body will be totally opaque

Q6. According to Newton's law of cooling the rate of cooling of a body is proportional to:-
(A) temperature of the body
(B) temperature of the surrounding
(C) fourth power of the temperature of body
(D) difference of the temp. of the body and the surrounding.

Q7. The adjoining diagram shows the spectral energy density distribution E_λ of a black body at two different temperatures. If area under the curves are in the ratio 16:1, the value of temp T is:

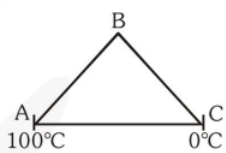


(A) 2000 K (B) 4000 K
(C) 8000 K (D) 6000 K

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q8. Three rods of equal length of same material are joined to form an equilateral triangle ABC as shown in figure. Area of cross–section of rod AB is S, of rod BC is 2S and that of AC is S, then



Column I	Column II
(A) Temperature of junction B	(P) Greater than 50°C
(B) Heat current in AB	(Q) Less than 50°C
(C) Heat current in BC	(R) Is equal to heat current in BC
	(S) Is $\frac{2}{3}$ times heat current in AC
	(T) None

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q9. A wall has two layer A and B each made of different material, both the layers have the same thickness. The thermal conductivity of the material A is twice that of B. Under thermal equilibrium the temperature difference across the wall B is 36°C. The temperature difference across the wall A is-

SUBJECTIVE TYPE

Q10. Two spherical ideal black bodies of radii r_1 and r_2 and with surface temperature T_1 and T_2 respectively, If both radiate the same power. Then calculate the ratio of $\frac{T_1}{T_2}$ is.

Q11. One end of a 2.35m long and 2.0cm radius aluminium rod ($K = 235\ W.m^{-1}K^{-1}$) is held at 20^0C. The other end of the rod is in contact with a block of ice at its melting point. The rate in $kg.s^{-1}$ at which ice melts is

[Take latent heat of fusion for ice as $\frac{10}{3}\times10^5\ J.kg^{-1}$]

Q12. Cooling rate of a sphere of 600 K at external environment (200 K) is R when the temperature of sphere is reduced to 400 K than cooling rate of the sphere is.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. In which process, the rate of transfer of heat is maximum :–

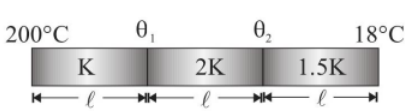
(A) conduction

(B) convection

(C) radiation

(D) in all these, heat is transferred with the same velocity

Q2. Calculate $(\theta_1+\theta_2)$ in shown situation.



(A) 150°C (B) 120°C

(C) 100°C (D) 190°C

Q3. Two walls of thicknesses d_1 and d_2 and thermal conductivity K_1 and K_2 are in contact. In the steady state, if the temperatures at the outer surface are T_1 and T_2, the temperature at the common wall is-

(A) $\frac{K_1T_1d_2+K_2T_2d_1}{K_1d_2+K_2d_1}$

(B) $\frac{K_1T_1+K_2T_2}{d_1+d_2}$

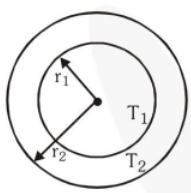
(C) $\left(\frac{K_1d_1+K_2d_2}{T_1+T_2}\right)T_1T_2$

(D) $\frac{K_1d_1T_1+K_2d_2T_2}{K_1d_1+K_2d_2}$

Q4. Two Plates of equal areas are placed in contact with each other. Their thickness are 2cm and 3cm respectively. Temperature of external surface of first plate is –25° C and that of external surface of second plate is 25° C What will be temperature of contact surface if the plates :-

(i) Are of same material
(ii) Have thermal Conductivity in ratio 2 : 3.
(A) (i) –5°C (ii) 0°C
(B) (i) 5°C (ii) 0°C
(C) (i) 0°C (ii) –5°C
(D) None of these

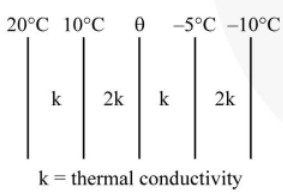
Q5. The figure shows a system of two concentric spheres of radii r_1 and r_2 and kept at temperatures T_1 and T_2, respectively. The radial rate of flow of heat in a substance between the two concentric spheres, is proportional to :-



(A) $\frac{(r_2 - r_1)}{(r_1 r_2)}$ (B) $\ell n\left(\frac{r_2}{r_1}\right)$

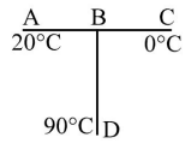
(C) $\frac{r_1 r_2}{(r_2 - r_{1)}}$ (D) $(r_2 - r_1)$

Q6. The figure shows the face and interface temperature of a composite slab containing of four layers of two materials having identical thickness. Under steady state condition, find the value of temperature θ.



(A) 5°C (B) 10°C
(C) –15°C (D) 15°C

Q7. Three conducting rods of same material and cross-section are shown in figure. Temperature of A, D and C are maintained at 20°C, 90°C and 0°C. Find the ratio of length BD and BC if there is no heat flow in AB

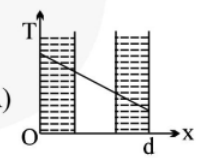


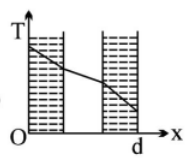
(A) $\frac{7}{2}$ (B) $\frac{3}{2}$ (C) $\frac{9}{2}$ (D) $\frac{5}{2}$

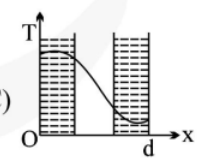
Q8. The power radiated by a black body is P and it radiates maximum energy around the wavelength λ_0. If the temperature of the black body is now changed so that it radiates maximum energy around wavelength $3/4\lambda_0$, the power radiated by it will increase by a factor of :-

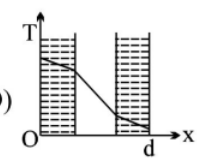
(A) 4/3 (B) 16/9
(C) 64/27 (D) 256/81

Q9. The wall with a cavity consists of two layers of brick separated by a layer of air. All three layers have the same thickness and the thermal conductivity of the brick is much greater than that of air. The left layer is at a higher temperature than the right layer and steady state condition exists. Which of the following graphs predicts correctly the variation of temperature T with distance d inside the cavity?

(A) 

(B) 

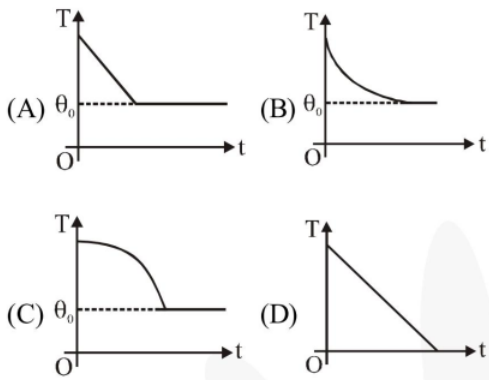
(C) 

(D) 

Q10. Water of volume 2L in a container is heated with a coil of 1 kW at 27°C. The lid of the container is open and energy dissipates at rate of 160 J/s. In how much time temperature will rise from 27°C to 77°C ? (Give specific heat of water is 4.2 kJ/kg) :-

(A) 8 min 20 s (B) 6 min 2 s
(C) 7 min (D) 14 min

Q11. If a piece of metal is heated to temperature θ and then allowed to cool in a room which is at temperature θ_0, the graph between the temperature T of the metal and time t will be closest to :



Q12. Spheres P and Q are uniformly constructed from the same material which is a good conductor of heat and the radius of Q is thrice the radius of P. The rate of fall of temperature of P is x times that of Q when both are at the same surface temperature? The value of x is :

(A) 1/4 (B) 1/3

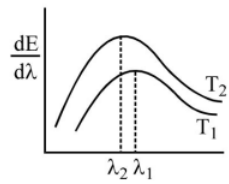
(C) 3 (D) 4

Q13. A system S receives heat continuously from an electrical heater of power 10W. The temperature of S becomes constant at 50°C when the surrounding temperature is 20°C. After the heater is switched off, S cools from 35.1°C to 34.9°C in 1 minute. The heat capacity of S is:–

(A) 100 J/°C (B) 300 J/°C

(C) 750 J/°C (D) 1500 J/°C

Q14. The spectral emissive power E_λ for a body at temperature T_1 is plotted against the wavelength and area under the curve is found to be A. At a different temperature T_2 the area is found to be 9A. Then λ_1/λ_2 =



(A) 3 (B) 1/3

(C) $1/\sqrt{3}$ (D) $\sqrt{3}$

Q15. A black body is at a temperature of 2880 K. The energy of radiation emitted by this object with wavelength between 499 nm and 500 nm is U_1, between 999 nm and 1000 nm is U_2 and between 1499 nm and 1500 nm is U_3. The Wien constant $b = 2.88 \times 10^6$ nm K. Then -

(A) $U_1 = 0$ (B) $U_3 = 0$

(C) $U_1 > U_2$ (D) $U_2 > U_1$

Q16. Three discs A, B, and C having radii 2 m, 4 m and 6 m respectively are coated with carbon black on their outer surfaces. The wavelengths corresponding to maximum intensity are 300 nm, 400 nm and 500 nm respectively. The power radiated by them are Q_A, Q_B and Q_C respectively.

(A) Q_A is maximum

(B) Q_B is maximum

(C) Q_C is maximum

(D) $Q_A = Q_B = Q_C$

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. In a 10–metre–deep lake, the bottom is at a constant temperature of 4°C. The air temperature is constant at –4°C. The thermal conductivity of ice is 3 times that of water. Neglecting the expansion of water on freezing, the maximum thickness of ice will be :-

(A) 7.5 m (B) 6 m

(C) 5 m (D) 2.5 m

Q2. The heat of flowing through a rod of length 50 cm and area of cross-section 5 cm^2. Its ends are respectively at 25°C and 125°C. The coefficient of thermal conductivity of the material of the rod is 0.092 kcal/m × s × C. The temperature gradient in the rod is :-

(A) 2°C/cm

(B) 0.2°C/m

(C) 2°C/m

(D) 20°C/m

Q3. It is hotter for the same distance over the top of a fire than it is in the side of it, mainly because :–

(A) air conducts heat upwards

(B) heat is radiated upwards

(C) convection takes more heat upwards

(D) convection, conduction and radiation all contribute significantly transferring heat upward

Q4. The emissive power of a black body at T = 300 K is 100 Watt/m^2. Consider a body B of area A = 10 m^2, coefficient of reflectivity r = 0.3 and coefficient of transmission t = 0.5. Its temperature is 300 K. Then which of the following is incorrect:–

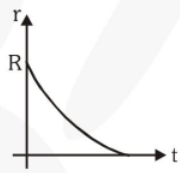
(A) The emissive power of B is 20 W/m^2

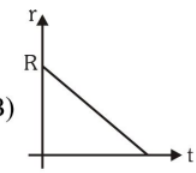
(B) The emissive power of B is 200 W/m^2

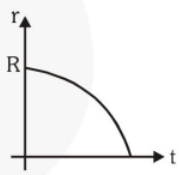
(C) The power emitted by B is 200 Watt

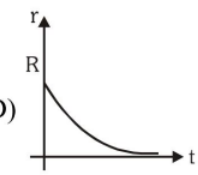
(D) The emissivity of B is = 0.2

Q5. A sphere of ice at 0^0C having initial radius R is placed in an environment having ambient temperature > 0^0C. The ice melts uniformly, such that shape remains spherical. After a time 't' the radius of the sphere has reduced to r. Which graph best depicts r(t)

(A) 

(B) 

(C) 

(D) 

Q6. A hollow copper sphere & a hollow copper cube of same surface area & negligible thickness, are filled with warm water of same temperature and placed in an enclosure of constant temperature, a few degrees below that of the bodies. Then in the beginning :-

(A) The rate of energy lost by the sphere is greater than that by the cube

(B) The rate of energy lost by the two are equal

(C) The rate of energy lost by the sphere is less than that by the cube

(D) The rate of fall of temperature for sphere is less than that for the cube.

Q7. 'Gulab Jamuns' (assumed to be spherical) are to be heated in an oven. They are available in two sizes, one twice bigger (in radius) than the other. Pizzas (assumed to be discs) are also to be heated in oven. They are also in two sizes, one twice big (in radius) than the other. All four are put together to be heated to oven temperature. Choose the correct option from the following:

(A) Both size gulab jamuns will get heated in the same time.

(B) Smaller gulab jamuns are heated before bigger ones.

(C) Smaller pizzas are heated before bigger ones.

(D) Bigger pizzas are heated before smaller ones.

Q8. An ideal black–body at room temperature is thrown into a furnance. It is observed that

(A) initially it is the darkest body and at later times the brightest

(B) it is the darkest body at all times

(C) it cannot be distinguished at all times

(D) initially it is the darkest body and at later times it cannot be distinguished

Q9. A long solid cylinder is radiating power. It is remoulded into a number of smaller cylinders, each of which has the same length as original cylinder. Each small cylinder has the same temperature as the original cylinder. The total radiant power emitted by the pieces is twice that emitted by the original cylinder. How many smaller cylinders are there ? Neglect the energy emitted by the flat faces of cylinder.

(A) 3 (B) 4

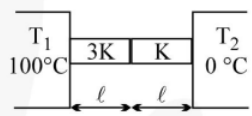
(C) 5 (D) 6

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 10 to 12

Two rods A and B of same cross-sectional are A and length l connected in series between a source (T_1 = 100°C) and a sink (T_2 = 0°C) as shown in figure. The rod is laterally insulated.



Q10. The ratio of the thermal resistance of the rods is -

(A) $\frac{R_A}{R_B} = \frac{1}{3}$ (B) $\frac{R_A}{R_B} = 3$

(C) $\frac{R_A}{R_B} = \frac{3}{4}$ (D) $\frac{4}{3}$

Q11. If T_A and T_B are the temperature drops across the rod A and B, then -

(A) $\frac{T_A}{T_B} = \frac{3}{1}$ (B) $\frac{T_A}{T_B} = \frac{1}{3}$

(C) $\frac{T_A}{T_B} = \frac{3}{4}$ (D) $\frac{T_A}{T_B} = \frac{4}{3}$

Q12. If G_A and G_B are the temperature gradients across the rod A and B, then -

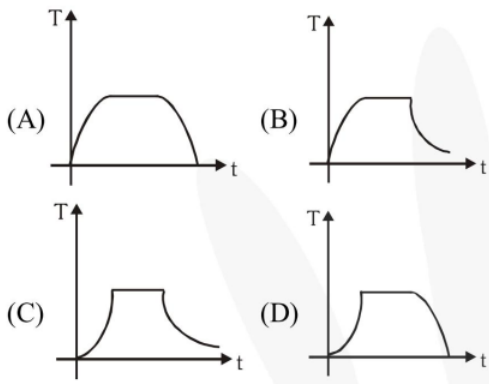
(A) $\frac{G_A}{G_B} = \frac{3}{1}$ (B) $\frac{G_A}{G_B} = \frac{1}{3}$

(C) $\frac{G_A}{G_B} = \frac{3}{4}$ (D) $\frac{G_A}{G_B} = \frac{4}{3}$

Paragraph for Question No. 13 to 15

In a home experiment, Ram brings a new electric kettle with unknown power rating. He puts 1 litre water in the kettle and switches on. But to his dismay, the temperature becomes constants at 60^0C after some time. The room temperature is 20^0C. Ram gets bored and switches off the kettle. He sees that during first 20 s water cools down by 2^0C.

Q13. Which is the best graph for temperature v/s time?



Q14. What is the wattage of the kettle

(A) 840W (B) 1W

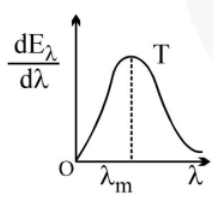
(C) 100W (D) 420 W

Q15. What is the time taken for the water to cool to 40^0C. (Approx)

(A) 510 s (B) 270 s (C) 120 s (D) 410 s

Paragraph for Question No. 16 to 17

The figure shows a radiant energy spectrum graph for a black body at a temperature T.



Q16. Choose the **CORRECT** statement(s)

(A) The radiant energy is not equally distributed among all the possible wavelengths.

(B) For a particular wavelength the spectral intensity is maximum.

(C) The area under the curve is equal to the total rate at which heat is radiated by the body at that temperature.

(D) None of these

Q17. If the temperature of the body is raised to a higher temperature T', then choose the correct statement(s)

(A) The intensity of radiation for every wavelength increases

(B) The maximum intensity occurs at a shorter wavelength

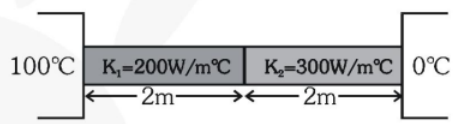
(C) The area under the graph increases

(D) The area under the graph is proportional to the fourth power of temperature

NUMERICAL TYPE

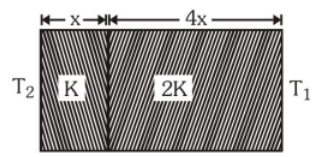
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q18. For shown situation, calculate the temperature of the interface.



SUBJECTIVE TYPE

Q19. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and 2K and thickness x and 4x, respectively are T_2 and T_1 ($T_2 > T_1$). The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2 - T_1)K}{x}\right)$ f, with f equals to-



Q20. If a liquid takes 30 sec. in cooling of 80°C to 70°C and 70 sec in cooling 60°C to 50°C, then find the room temperature.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	B	B	A	A	A	C	D	A	A
11	12	13	14	15	16				
B	D	D	A	A	B				

Ex-1A

1. B **2.** C **3.** A **4.** A **5.** D

6. D **7.** B **8.** (A)-P; (B)-R; (C)-T **9.** 18°C

10. $\sqrt{\frac{r_2}{r_1}}$ **11.** $2.4\pi \times 10^{-6}$ **12.** $\frac{3}{16}$ R

Ex-2									
1	2	3	4	5	6	7	8	9	10
C	D	A	A	C	A	A	D	D	A
11	12	13	14	15	16				
B	C	D	D	D	B				

Ex-2A

1. A **2.** A **3.** C **4.** A **5.** B

6. B,D **7.** B **8.** D **9.** B **10.** A

11. B **12.** B **13.** B **14.** D **15.** B

16. A,B **17.** A,B,C,D **18.** 40°C **19.** 1/3 **20.** 40°C

-: Important Notes & Formulas :-